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Original Research

Product-Service Systems for Furniture: A Literature Review to Redefine the Relationship with Sustainability

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Abstract: The furniture industry represents a significant sector for the Italian and European economies. This sector is currently facing numerous environmental challenges and undergoing a transition toward sustainability. The concept of Product-Service Systems (PSS) offers a potential strategy to address this change by generating sustainable values. This article aims to analyze and present the state of the art in scientific studies on the development of PSS as an approach to achieve sustainability and circularity in the furniture sector. Through a literature review, twenty-three contributions were selected and analyzed, revealing unexplored sources and areas for future studies on the topic, such as national-specific features, cultural dimensions, and stakeholder experiences, including those of users and actors in the value chain. In addition, at present, there is a strong focus on specific product areas and on product-oriented and use-oriented PSS. The findings highlight the potential and necessity of leveraging scientific resources to implement new PSS solutions and generate valuable learning experiences.

Keywords: *Design for Sustainability (DfS), Circular Economy, Product-Service Systems (PSS), Servitization, Furniture*

Introduction

The Challenges and Opportunities of Sustainability in the Furniture Sector

The furniture industry holds strategic importance for the European Union (EU), with one-third of the world's marketed furniture coming from the Old Continent (JRC 2023). Today, this sector faces several challenges related to resource exploitation and the systemic impacts of overproduction and consumption. Annually, 10.5 million tons of furniture are produced, while 10 million tons are discarded—most of which before the end of their useful life (Forrest et al. 2017). A small percentage is reused, with 80% to 90% incinerated or sent to landfills, constituting about 4% of municipal solid waste in the EU (Forrest et al. 2017). This waste is largely caused by trends and changing fashions, leading users to change their furniture more frequently, shortening their useful life (Centre for Design at RMIT 2001; Conlon 2019). Some Life Cycle Assessment (LCA) analyses show (Iritani et al. 2015; Medeiros et al. 2017; Sakib et al. 2024) that the most impactful life cycle stages are pre-production (material extraction and

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processing) and disposal due to the intensive use of raw material, adhesives, dyes, coatings, and the release of volatile organic compounds (Azizi et al. 2016; Lewis et al. 2017). Also, transportation has a significant environmental impact, whether in distributing furniture to retailers or delivering it to consumers' homes.

In this sense, the furniture sector—like other industrial sectors—shows the limits of a system and society based on finite resources, growing demand, and an increasingly short and linear product life cycle (based on take–make–dispose). Therefore, it becomes a priority to understand how the concept of sustainable development can also be integrated into this sector. Sustainable development was defined in 1987 by the Bruntland Commission as "a development that meets the needs of the present without compromising the ability of future generations to meet their own needs" (WCED 1987). Although this definition is noteworthy since it was the first of its kind, it has been regarded more as a slogan than as the basis for a theory (Banerjee 2008). This is because it starts from an anthropocentric view, putting the well-being of future human generations at the center, and not of the entire ecosystem (Bistagnino 2009). Because of this, the sustainable development concept was further explored by various authors and organizations, such as UNESCO, which—in line with what had previously been done by IUCN, UNEP, and WWF (1991)—emphasized the need for diversity, both for nature (through biodiversity) and for humans (through cultures) to support growth, not strictly linked to economic growth. Also, Costanza and Patten (1995) integrate this view of biological development with economic development, emphasizing the temporal aspect of sustainability.

Among the various theories developed to link the different dimensions of sustainable development to make it more operational, one that has gained the most traction is the circular economy (CE). CE builds on Industrial Ecology (IE), cradle-to-cradle, and performance economics and has been defined by the Ellen McArthur Foundation (2013) as "an industrial economy that is restorative or regenerative by intention and design." CE aims to keep the functional use of products and materials within the economic sphere for as long as possible by creating closed loops of material flows to reduce material and energy inputs and keep the waste within the production system (Mallick et al. 2023; Ceschin and Gaziulusosy 2019). To facilitate the understanding of the potential of retaining the value of materials or products in the life cycle, the R-hierarchy was defined. The R's have been referred to as the key to product sustainability and for holistic considerations of material flows in a product life cycle (Go et al. 2015). Initially, the hierarchy started with the 3Rs (Reduce, Reuse, and Recycle). However, over time, more were added: Jawahir et al. (2005) included three more (Recover, Redesign, Remanufacture); Reike et al. (2018) summarized four more (Refuse, Repair, Refurbish, and Re-mine) for a total of 10R's. Several strategies have been developed to implement CE to transform waste outputs into valuable input resources, such as IE, Cleaner Production, Industrial Symbiosis, Product-Service Systems (PSS), and EPR (Campbell-Johnston et al. 2020). These strategies tie in with the R-hierarchy and make it more operational, trying to

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keep resources in the system for as long as possible while keeping their value high, and simultaneously trying to decouple the economic success of companies from the amount of resources used. Although these strategies mainly focus on resource flow, they have wide-ranging impacts, changing enterprises' products, systems, and business models (Burke et al. 2023). Nevertheless, if implemented, the strategies have the potential to align the interests of companies and systems with the concepts of sustainable development and CE.

To introduce the concepts of sustainable development and CE in the furniture sector, two main strategies have been proposed in the current literature: designing furniture to facilitate the recycling of materials and extending their useful life to decrease consumption (e.g., Reuse, Repair, Refurbish) (Besch 2005; Witte 2000). In the first case, disassembly and proper recycling are not always cost-effective practices (as they are highly dependent on available product quantities and market fluctuations) or easily applicable (as they are dependent on previous design and marking of individual materials) (Witte 2000). Although, to date, companies are trying to minimize material use and select resources with a low impact (Iritani et al. 2015; Rinawati et al. 2018), this is still quite insufficient. In the second case, several (design) strategies are adopted to extend the life of a piece of furniture, such as design for durability, Repair, and maintenance (Loučanová et al. 2022). As shown by Pei et al. (2024), the furniture industry is product-centric by focusing on interventions at the material or product level and leaving BM, PSS, or Business Ecosystem level still to be explored. Indeed, these strategies can be implemented beyond new product designs through innovative services and circular or sustainable business models (Lüdeke-Freund et al. 2019).

The Product Service Systems and Sustainable Development

PSS—the integration of products and services—offer a viable solution for emissions reduction. Academic literature identifies three types of PSS (Tukker 2004; Tukker and Tischner 2006):

- Product-oriented: the sale of products is the focus of the business model, and services are eventually integrated around this.
- Use-oriented: the product and its sale are no longer the driving force of the business model. The company retains possession of the products and provides access to them in various formats and time frames, subject to payment.
- Result-oriented: the customer and the company agree on a result and the products involved are not predefined, also requiring reduced end-user involvement.

PSS increase the intensity of product use, enabling shared utilization by multiple users, thereby reducing resource consumption and costs, particularly for raw materials. Moreover, they enhance product value and economic returns for companies. Indeed, Goedkoop et al. (1999) and Tukker (2015) argue that PSS can decouple economic growth from material

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consumption by creating revenue through services rather than through product sales, enabling profit from a more efficient use of resources. Similarly, Stahel (2010) highlights that the performance economy can drive economic returns by promoting reuse, repair, and service-based contracts that reduce cost and resources, creating new employment opportunities. Oliva and Kallenberg (2003) emphasize that transitioning to service-based offerings allows firms to stabilize revenue streams and capture value beyond one-time product sales, deepening the customer relationship. However, Vezzoli et al. (2015) note that realizing the PSS potential depends on overcoming design, market, and infrastructural challenges.

PSS represent a viable way of doing business to align companies with their environmental responsibility for a sustainable society, to meet the principles of the sustainability while creating a new channel between entrepreneurs and consumers (Manzini 1996; Tukker 2015). Companies are increasingly experimenting with services and business models that implement or extend the product's life, such as repair services, remanufacturing, rental, and leasing. Also, institutions like the European Commission underscore the importance of this sector for European enterprises, initiating discussions on future legislation to promote furniture reuse (Yiapanis and Gehring 2022). Thus, investigating PSS to reduce resource use and foster a sustainable transition is a topic of strategic interest to various stakeholders.

The discipline of design has already been linked to sustainability and PSS. In fact, the link between design and sustainability evolved at the same time as the ecological movements of the 1970s and 1980s, generating various approaches, including green design and ecodesign (Vezzoli and Manzini 2008). These aim to minimize the use of the planet's resources and benefit users by broadening the designer's perspective (Ceschin and Gaziulusoy 2019). Over time, different approaches of design for sustainability (DfS) have been developed, broadening the view of design from individual components to products, to systems (Ceschin and Gaziulusoy 2019). Ceschin (2012) emphasizes the value of designing PSSs through a design approach, linking them directly to sustainability (S.PSSD) and stressing their potential to drive systemic innovation necessary for fostering sustainable transitions. Innovating a system means changing how a societal need is satisfied, simultaneously changing the consumption and production patterns of goods and services, as well as the products and services themselves (Schmidt-Bleek 1993). Designers possess the skills to tackle broad challenges like sectoral or organizational sustainable transitions. Indeed, in recent decades, the focus of design has shifted from individual products to services and systems. Consequently, the design and provision of PSS based on sustainability principles is a growing trend (Loučanová et al. 2021).

Currently, the available literature provides insights into specific types of PSS and their contextual applications (Sundin and Bras 2005, Ceschin 2012). However, general indications are not always valid due to the peculiarities of the furniture product, market, and industry, and it becomes necessary to develop sector-specific knowledge (Baldassarre et al. 2020). Indeed, although the challenge of transitioning to circularity is widespread, each industrial sector has its specific methods and issues, requiring individual approaches and careful analysis

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(Koszewska and Bielecki 2020). Furthermore, companies need clear guidance, with solutions tailored to specific product groups that are economically and technologically viable (Koszewska and Bielecki 2020). Nowadays, a comprehensive overview of how the furniture sector is addressing the challenges of a sustainable and circular transition through PSS is lacking. This gap is evident in terms of sector-specific strategies, application of design methods and stakeholder dynamics. Accordingly, this article aims to analyze the existing academic literature to map the state of the art of PSS designed for sustainable transition in this sector. Through a systematic literature review based on Scopus and ISI Web of Science, twenty-three contributions were analyzed to map the current state of knowledge on PSS for sustainability in the furniture sector. Doing so seeks to transform scientific studies into actionable insights, offering a solid foundation to support the design, implementation, and commercialization of circular and sustainable solutions while generating future research opportunities. Indeed, the insight that emerged from this research aims to go beyond theoretical understanding and inform the design, implementation, or adoption of PSS in the furniture sector, helping both academics and professionals.

In the following sections, the article introduces the methodology through which the letter review was conducted. Afterward, the results are presented, commented, and critically discussed, highlighting future research opportunities for researchers and areas of intervention for companies and practitioners.

Methodology

To outline the state of the art, a systematic literature review was conducted in three phases: planning, execution, and synthesis (Tranfield et al. 2003). The initial step defined the protocol, objectives, and keywords (in English) to be used in databases such as Scopus and ISI Web of Science (WoS). This preliminary exploration confirmed the research gap and refined the keywords. The selected terms were used in various combinations within the aforementioned databases during June and July 2024, targeting the title, keywords, and abstract.

Early results indicated that using sustainability-related keywords like “Circular Economy” and “Sustainability” was redundant, as most emerging articles were already centered on these topics. For instance, when these keywords were combined with the sector-related term (e.g., furniture or similar term), very few articles emerged, limiting the scope significantly. In contrast, a broader search not focusing on the furniture sector produced more results, yet a quick reading of the titles and abstracts of the articles revealed that most contributions analyzed PSS as a means to achieve sustainability or circularity. Thus, including keywords such as “Sustainability” or “Circular Economy” would not have added specificity; on the contrary, it would have diluted the focus on sector-specific contribution. The keyword definition highlights how the PSS theme is inherently tied to environmental benefits and the CE. However, previous studies have noted that implementing a PSS does not automatically

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improve sustainability (Tukker and Tischner 2006). To ensure international relevance, the review focused on English-language contributions without time restrictions. The finalized list of keywords and their combinations takes these considerations into account and is presented in Table 1.

Once the search criteria were defined in the selected databases, the second phase of the literature review process began: execution. The search yielded forty-five contributions for Scopus and sixty-seven contributions for WoS. These were mapped within a spreadsheet to schematize information such as title, authors, year of publication, keywords, and nationality of the first author. In addition, the spreadsheet was used to highlight and manage duplicate contributions. This process yielded ninety-eight unique contributions across both databases, which have been further refined according to this article's specific topics of interest.

Table 1: Keywords Used in the Research and Resulting Contributions in the Databases

Keywords	*Database*	
	Scopus	*ISI Web of Science*
Furniture OR Furniture Industry AND Product-Service System* OR PSS OR PSSD OR Servitization	45	67
Total Number of Unique Contributions	98	

Subsequently, the ninety-eight contributions were screened to identify those relevant to the research focus. The criteria for inclusion of articles in the literature review included examples of analysis, design, or implementation of PSS in the furniture sector. On the contrary, while the contribution focused on production processes, materials, PSS management processes, and software, communication of PSS, quantitative calculation of economic-productive aspects, macro-economic aspects, and legislation were excluded. The selection process included reading each contribution's title, keywords, and abstract. This process resulted in eighteen relevant texts, which were fully read to assess their relevance further, culminating in a final sample of seventeen contributions. Applying the snowballing technique—i.e., analyzing sources and citations—added six more contributions, bringing the final total to twenty-three. These include journal articles, book chapters, reports, and conference proceedings. The complete list of contributions is shown in Table A1, in the Appendix.

In the third phase, synthesis—including data extraction, analysis, and clustering—was carried out iteratively by the two authors, working independently to avoid cognitive biases. Documents were initially categorized by year, publication type, and the nationality of the first author. Using content analysis and, in particular, the SLIP method (Sort, Label, Integrate, and Prioritize key concepts) (Julien 2008; Maeda 2006), key themes were identified

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by searching for relevant terms within the texts, which were then grouped and clustered based on their associations. The purpose of this analysis is to delve into the specific contributions and aspects investigated by the selected studies to identify the most explored themes and extrapolate directions for future research. The contributions were then categorized according to five key aspects: article contributions, research objectives, types of furniture examined, types of PSS analyzed, and design strategies and/or business models adopted. The authors subsequently summarized and interpreted the findings, identifying the state of the art of PSS research in the furniture sector and highlighting future research and business opportunities. Figure 1 summarizes the literature review process.

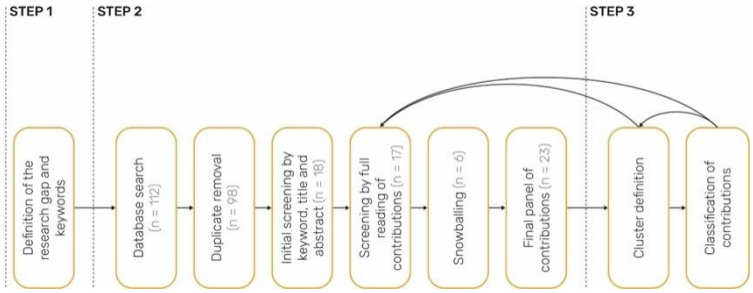

Figure 1: Process Adopted for the Systematic Literature Review

Results

Analytical Characterization of Contributions

The initial analysis examined the distribution of contributions by year, publication type, and nationality of the first author. Figure 2 shows the number of articles published per year. Among the twenty-three contributions, twenty were published after 2015, indicating a recent surge of interest in the topic in the furniture sector despite its emergence in the early 2000s. Peaks in publication activity occurred in 2017 and 2021. The temporal distribution of contributions reveals increased interest in the furniture sector from 2015 onward. This trend reflects two converging dynamics: on the one hand, the increasing maturity and diffusion of the PSS concept outside the disciplines in which it was developed, its hybridization with the design theme, and its application in various sectors; on the other hand, it reveals the growing importance of this concept in addressing new legislation and policy agendas during the same period (e.g., the EU Circular Economy Action Plan). In this sense, the recent EU's Ecodesign for Sustainable Product Regulation, which has prioritized furniture to integrate circularity principles in design, could provide a further boost.

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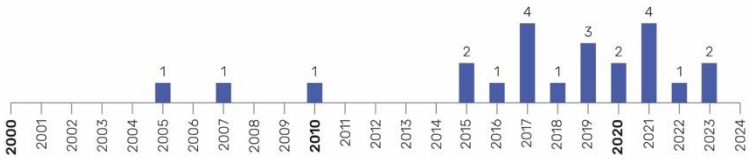

Figure 2: Historical Distribution of Published Contributions

A total of twelve contributions (nearly half of the sample) were published in scientific journals, eight in conference proceedings, two as book chapters, and one as a report (Figure 3). Among the journals, the most popular were the *Journal of Cleaner Production* (three publications) and *Sustainability (Switzerland)* (two publications). Only one contribution was published in a design-specific journal, namely *The Design Journal*. Regarding conferences, on the other hand, the majority of them have an engineering background, whether related to design engineering, industrial engineering, or management engineering (e.g., *CIRP* and *ICED*). A minority of them are specific conferences on PSS or wood.

The strong presence of contributions in the *Journal of Cleaner Production* and *Sustainability (Switzerland)* highlights how sustainability-oriented journals are leading the discourse on PSS, also in the furniture sector. This insight reinforces the keyword finding—i.e., PSS is often discussed as a strategy for sustainability. In contrast, although design plays a strategic role in developing PSS, the absence from design-related journals and conferences suggests that the design research community still has room to provide input.

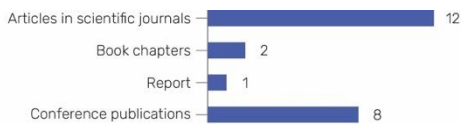

Figure 3: Distribution of Contributions by Publication Type

Regarding the first authors' nationality, the majority of contributions originate from European countries, particularly Northern Europe, followed by Southern Europe and Asia (Figure 4). Sweden is the leading contributor (five contributions), followed by Denmark (four contributions) and the Netherlands (three contributions). They are followed by Slovakia and South Korea with two contributions, and the United Kingdom, Portugal, Italy, Greece, Japan, Thailand, and Colombia with one contribution.

The geographical distribution of contributions shows a clear dominance of Northern European countries, with limited representation from Southern Europe and Asia. This imbalance may be due to several factors. Firstly, Northern Europe has pioneered sustainability-oriented research and practice, often supported by robust public, national, and European funding. An example may be the USE-IT-WISELEY (UIW) project that facilitated the creation and application of new business models (including PSS) (Bosch et al. 2017) or

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the more recent Nordic Circular Design Programme[1] developed in partnership with the Danish Design Centre.

In contrast, countries such as Italy—despite having a strong furniture industry and design heritage—have made minimal contributions to PSS in furniture, probably due to fewer specialized funds and the primary importance of product sales in companies' business models. From these observations, one insight gained is the opportunity for future research to explore drivers and barriers at a national level to have a greater geographical diversification of contributions besides being sector-specific.

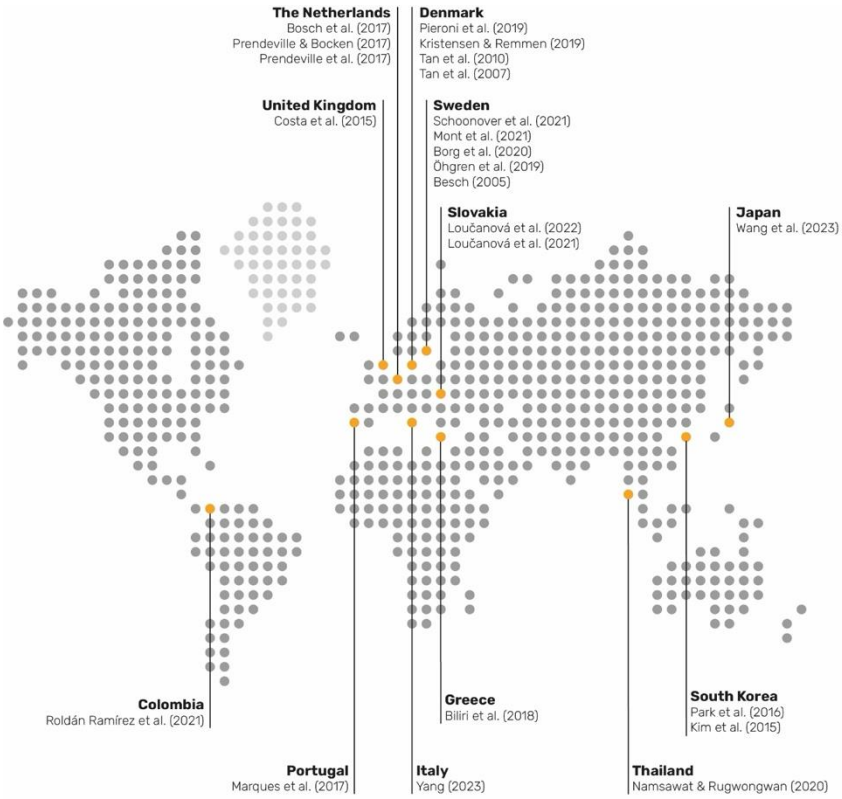

Figure 4: Geographic Distribution of Contributions by the Nationality of the First Author

Thematic Analysis of Contributions

As anticipated, further categorization was developed to understand the PSS literature in the furniture sector. In the following paragraphs, the five clusters of interest (contribution of the article, research objective, types of furniture under consideration, types of PSS under consideration, design strategies, and/or business models adopted) are reported and analyzed.

[1] Nordic Circular Design Programme: https://www.circulardesignprogramme.com/nordic/apply.

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Firstly, the articles can be distinguished based on their methodologies and the type of contributions they provide (Figure 5). On the one hand, some articles generate theoretical contributions through literature reviews, surveys, or interviews. On the other hand, others report field-based experiences or practical applications, such as case studies or action research. Nine articles contribute to the theory, while another nine focus on applying new methods and tools through case studies, often looking at the furniture sector as an application domain—e.g. Pieroni et al. (2019), Kristensen and Remmen (2019), and Park et al. (2016). The remaining five articles report action research conducted with companies in the sector. As shown in Figure 5, contributions from the latter category predominantly appeared before 2018, possibly due to the impact of the COVID-19 pandemic and the extended timelines required for this type of investigation (see Table A2 in the Appendix for an overview of the duration and partnerships of these contributions). In this regard, it is possible to point out how action research is linked to the work of individual authors—such as Dr. Prendeville—probably due to their network (Prendeville and Bocken 2017; Prendeville et al. 2017; Costa et al. 2015).

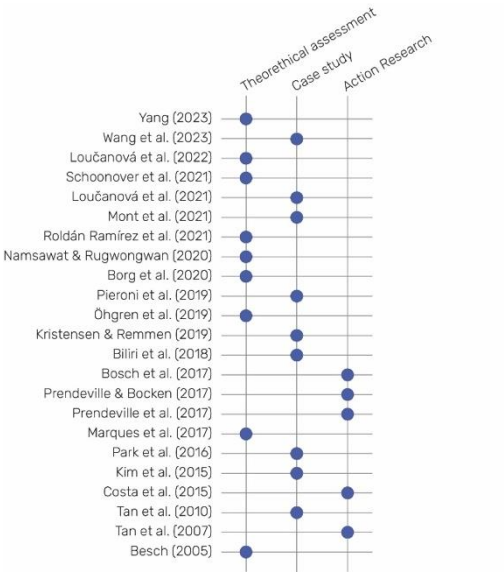

Figure 5: Types of Scientific Contributions in the Reviewed Articles

In terms of research objectives, the distribution across categories is diverse (Figure 6). On the total:

- Ten contributions focus on defining and testing design processes of a PSS, corresponding to the typical steps of a design process—i.e., an initial phase of exploratory research and context analysis (with visions and future scenarios, analysis of users and stakeholders, their needs, fears, and opportunities), followed by problem

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definition, concept generation, and development (Costa et al. 2015; Pieroni et al. 2019), and eventually testing and implementation actions (Kim et al. 2015; Wang et al. 2023).

- Seven articles present methods and tools to support the design and development of a PSS. A great variety emerges, which can be used at different stages and for different purposes, focusing on the user, the company, the business model, or the product. Among the most discussed are:
    - System maps (Costa et al. 2015; Pieroni et al. 2019; Wang et al. 2023)—i.e., visual representation of actors, processes, and relationships involved in delivering a PSS, used to understand complexity (Tassi 2019).
    - Personas (Costa et al. 2015; Wang et al. 2023)—i.e., fictional profiles representing key user archetypes based on real data, used to align solutions with users' needs, expectations, and behaviors (Tassi 2019).
    - Customer journeys (Costa et al. 2015; Kim et al. 2015; Pieroni et al. 2019)—i.e., chronological visualizations of a user's interactions with the PSS, to identify pain points, service gaps, and design opportunities (Tassi 2019).
    - Service blueprint (Costa et al. 2015; Kim et al. 2015)—i.e., layered diagrams mapping frontstage (user-facing) and backstage (organizational) activities of service, clarifying how services are delivered and coordinated internally (Tassi 2019).
    - Sustainable Business Model Canvas (Kim et al. 2015; Pieroni et al. 2019)—i.e., an adapted version of the traditional Business Model Canvas that integrates environmental and social dimensions into the core value proposition, revenue, and delivery logic (Osterwalder and Pigneur 2010).

  Less frequent ones are:

    - Scenario-building and backcasting techniques (Wang et al. 2023)—i.e., a method to envision desirable futures and plan backward to identify the steps needed to reach them, often used in long-term sustainability planning (Italia 2024);
    - LCA and variants (Costa et al. 2015)—i.e., a systematic analysis of the environmental impacts associated with all stages of a product or service life cycle, from raw materials to disposal (Glavič and Lukman 2007).

  In some articles, instead, methods and tools designed ad hoc for more specific aspects are presented, also leading to mixtures between the two (Biliri et al. 2018; Bosch et al. 2017; Kristensen and Remmen 2019; Marques et al. 2017; Park et al. 2016). Among the most significant, we can highlight the *Sentiment Analysis & Knowledge Management Tool* by Biliri et al. (2018), which aims to analyze user feedback or usage

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data to extract preferences and emotional responses to help companies adapt their PSS offering to user satisfaction and usage patterns. The *System Evolution and Concept Generation Tool* by Park et al. (2016) is also relevant, as it combines market trends with user needs to generate new PSS concepts to support companies in the sector with more complex models such as subscription or refurbishment models.

- Six contributions identify barriers and enablers of developing and implementing a PSS, which are analyzed both from the company's point of view (for implementation and design) (Mont et al. 2021; Schoonover et al. 2021) and from the user's point of view (Borg et al. 2020). Overall, seven recurring themes can be highlighted: new capabilities, financial and economic risks, product design and technical aspects, value chain relationships, consumer-related barriers, organizational paradigm shifts, and legislation (Besch 2005; Mont et al. 2021; Öhgren et al. 2019; Prendeville and Bocken 2017; Schoonover et al. 2021; Adrian Ronald Tan et al. 2007). In some cases, the articles also present solutions to overcome the emerging barriers. For example, Öhgren et al. (2019) recommend establishing large contracts that bundle big volumes across public agencies and include leasing and take-back schemes, reducing the financial risk for the service providers. Instead, Prendeville and Bocken (2017) propose that local manufacturing networks should be developed based on Design for Disassembly to overcome barriers related to logistical complexity, centralization, and end-of-life recovery. Nevertheless, companies and researchers often do not test these enablers in the published articles (Öhgren et al. 2019). When applied, they can be considered contextual propositions as they respond to specific contexts, and scalability needs to be verified by subsequent research (Prendeville and Bocken 2017).
- Seven contributions report transition experiences. These often adopt a holistic perspective, encompassing multiple aspects in the narrative (such as product, service, and business model), describing how product design choices have led to organizational and business model changes over time or vice versa. However, transition experiences often focus on a business perspective, describing the shift from product-centric to service-oriented models while overlooking the potential impact from and to the sociotechnical system and its interconnections (Benn and Kearins 2011). It should be emphasized that the process of change can also be observed in some cases through the conjunction of multiple contributions, tracing the path of the company in collaboration with the authors (Costa et al. 2015; Prendeville et al. 2017; Prendeville and Bocken 2017; Tan et al. 2007; Tan et al. 2010). In the case of this cluster, it is very difficult to draw common elements since each company and article presents specificities guided by strategic choices, possibilities of the organization, and the context in which it operates.

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- Three articles examine user perceptions from a national perspective (Slovakia, Sweden, and Thailand). Although each contribution analyses the topic with a different perspective, recurrences can be highlighted. Consumers are interested in product features, such as durability, materials, and recyclability. Furthermore, it is noted how even if services are not a mandatory element, they positively influence satisfaction, especially in post-sale and end-of-life phases (such as assembly, disassembly, take-back, and second-hand) (Borg et al. 2020; Loučanová et al. 2022; Namsawat and Rugwongwan 2020).

The analysis of results also revealed correlations between types of contributions and research objectives. Articles focusing on understanding consumer perceptions and acceptance levels typically contribute to theory through surveys (Loučanová et al. 2022; Namsawat and Rugwongwan 2020) or semi-structured interviews (Borg, Mont, and Schoonover 2020). Similarly, analyses of barriers and enablers follow a similar pattern. On the other hand, the definition and implementation of design methods and tools are often based on case studies. Transition processes are predominantly reported as field experiences—usually through action research—involving manufacturing companies (Bosch et al. 2017; Prendeville and Bocken 2017; Prendeville et al. 2017).

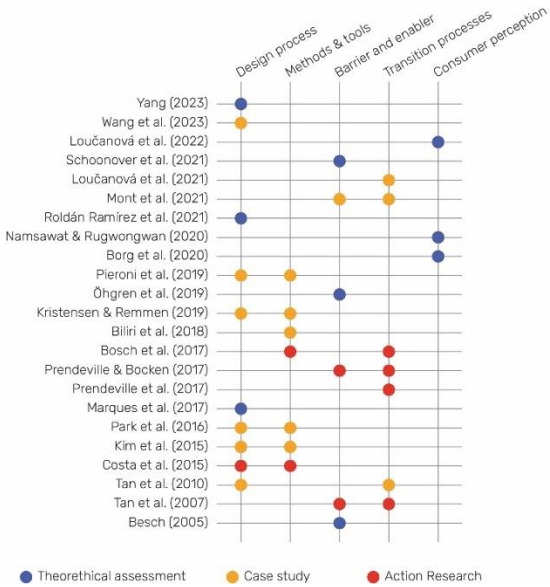

Figure 6: Research Focus and Objectives Compared to the Three Previously Identified Categories of Contributions

Focusing on the analyzed sector (Figure 7), the most studied furniture type is office furniture (fifteen out of twenty-three articles), followed by home furniture (six out of twenty-

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three). Four contributions do not specify a particular furniture sector, while isolated cases examine outdoor, school, or retail furniture. Historically, the interest in home furniture within PSS only emerged after 2019, while office furniture previously dominated the research focus. The prominence of office furniture can be attributed to its connection with the contract furniture sector, which is well-suited to business models such as renting, leasing, or remanufacturing, given its high production volumes, business-to-business (B2B) relationships, and supplementary services. This particular focus reflects the early development of participatory design in Scandinavia, where office furniture was identified as the subject of the first co-design experiments involving users, workers, and designers (Broberg 2010). Office furniture has proven to be a widely accepted choice over time, as it represents a structured, institutional, and shared setting in which organizations can easily experiment, engaging multiple stakeholders in the design process. Nevertheless, the COVID-19 pandemic and the resulting restrictions also caused an increased interest in wellness in private homes.

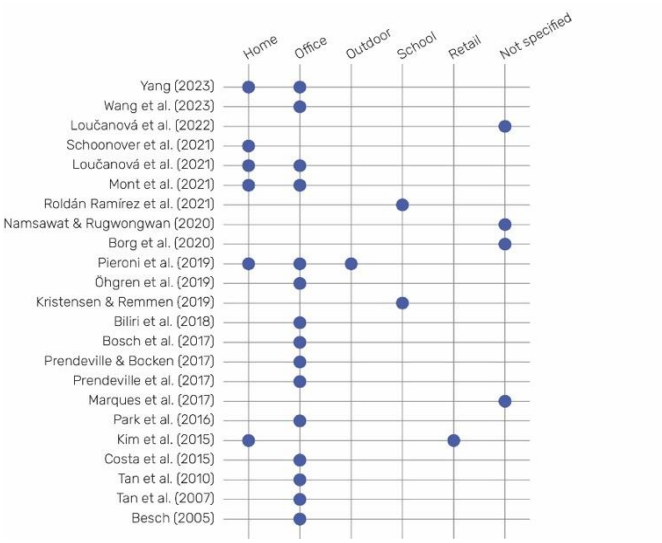

Figure 7: Types of Furniture Examined in the Analyzed Contributions

Literature discussed different types of PSS following Tukker's (2004) categories (Figure 8). Among all the articles, use-oriented PSS are the most analyzed (eleven out of twenty-three), followed by product-oriented PSS (six out of twenty-three). None of the contributions address result-oriented PSS, highlighting the gradual transition companies undergo and the product's importance in the furniture sector (Prendeville et al. 2017). Eight articles do not specify a particular type of PSS.

The predominance of use-oriented PSS, followed by product-oriented models, is linked to what was previously noted regarding the prevalence of office furniture—namely, the compatibility of contract furniture practices (e.g., leasing, renting) with the B2B sector and public institutions (Besch 2005; Öhgren et al. 2019; Mont et al. 2021). Moreover, these PSS

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models can be considered self-reinforcing, acting as enablers for circularity strategies such as remanufacturing or component reuse since they facilitate product recovery (Pieroni et al. 2019).

Conversely, the absence of result-oriented PSS may indicate a practical and theoretical gap. These models entail significant challenges regarding value perception, trust among stakeholders, and the reliability of the service infrastructure (Tukker and Tischner 2006; Vezzoli et al. 2015). Nevertheless, they represent a promising area for innovation in contexts such as co-working spaces, elderly care, or educational environments, where it is possible to establish B2B relationships based on achieved outcomes rather than individual products (e.g., workstation productivity, number or usage of workstations) (Ceschin and Gaziulusoy 2019). Moreover, result-oriented PSSs have lower environmental and resource costs than product- and use-oriented PSSs and are worth exploring further (Sarasini et al. 2024).

Finally, the fact that a substantial portion of the articles does not explicitly state the type of PSS explored may suggest a difficulty in classification. Future studies could better contextualize the categories proposed by Tukker (2004) within the furniture sector—thus helping companies and researchers position themselves in one category or another—and simultaneously facilitate categorization by introducing intermediate categories and enhancing permeability between them.

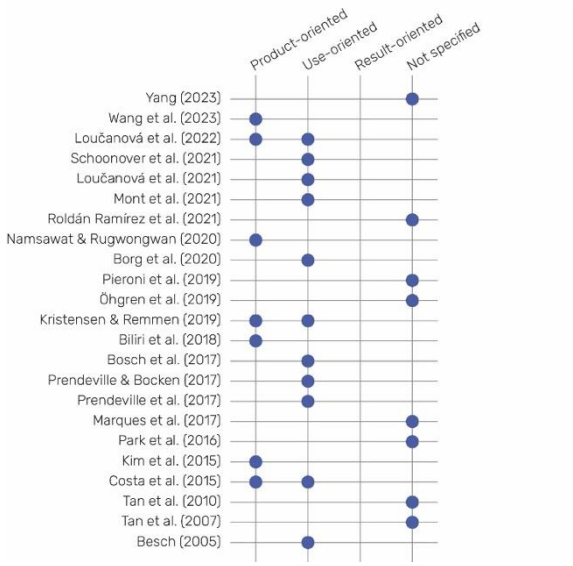

Figure 8: Types of PSS Analyzed by the Articles. The Categories Reflect Those Formalized by Tukker (2004)

Finally, various strategies are presented for achieving sustainability objectives through PSS, often with multiple strategies explored within a single article (Figure 9). Notably, most contributions do not specify a particular strategy (eleven out of twenty-three). Six contributions focus on remanufacturing and refurbishment, analyzing related experiences,

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barriers, and enablers. Although these two strategies differ, they are often studied together due to their similar logistical and organizational requirements. Five contributions address renting/leasing strategies, where the authors note that practitioners often use these terms interchangeably. Despite academic distinctions, they are frequently combined (Borg et al. 2020; Schoonover et al. 2021). A few articles explore upgrading, reuse, repair, and maintenance strategies.

Over time, the perception of renting/leasing and remanufacturing strategies has evolved. Early 2000s articles labeled these business models as economically unsustainable (Besch 2005). More recent contributions highlight successful practices in specific sectors and contexts (such as the transport sector in urban environments), shedding light on success factors (Borg, Mont, and Schoonover 2020; Mont, Lehner, and Schoonover 2021). This evolution may stem from sociocultural and legislative changes, as well as the emergence of new organizational and logistics systems (Bosch et al. 2017; Pieroni et al. 2019; Schoonover et al. 2021).

These findings can also be interpreted through established theoretical frameworks in the literature. Considering the framework by Bocken et al. (2016), the majority of the articles fall within the category of Business Model Strategies aimed at slowing resource loops—particularly through *Access and Performance Models* (which seek to meet user needs without requiring product ownership) and *Extending Product Value* (which focuses on exploiting the residual value of products). These strategies include refurbishment/remanufacturing, leasing/renting, and reuse. On the other hand, repair and maintenance fall under the *Classic Long-Life Model*, which is based on products with extended lifespans supported by design for durability and repairability. From this perspective, it emerges that the contributions in the literature aim to expand the business models within the furniture sector while overlooking aspects in which the sector and its products are already strong, i.e. long product lifespans. Similarly, comparing the results with the R-hierarchy proposed by Reike et al. (2018), it emerges that most articles and strategies correspond to mid-level imperatives, such as R4 (reuse), R5 (repair), R6 (refurbish), and R7 (remanufacture), aiming to extend product life or optimize resource use throughout the life cycle. Instead, renting and leasing align with higher-level strategies such as R0 (refuse) and R1 (reduce), based on product-sharing models rather than ownership.

Finally, when cross-referencing strategies with PSS types, a diverse distribution emerges. As seen in Figure 9, product-oriented and use-oriented PSS are evenly distributed across strategies, with no significant predominance. However, it may be interesting to point out that the use-oriented PSS mainly involves remanufacturing/refurbishment and leasing/renting strategies. From the connection with frameworks and considering the evolution of the perception of PSS's environmental and economic effectiveness, the focus on use-oriented PSS may be due to these models' recent maturity and feasibility. Use-oriented PSS allow the retention of ownership of the product, a crucial element in controlling the product life cycle and recovering the economic value at the end of life. As mentioned above, these models align

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with B2B contract furniture contexts (such as offices, public procurement, and public spaces) where long-term contracts and centralized logistics can facilitate take-back, remanufacturing, and service provision.

These considerations are also valid from a business model perspective, as it emerges that the experience and expectations of furniture users do not want to be affected while still trying to extract value and make resources efficient.

Therefore, the dominance of refurbishment/remanufacturing and leasing/renting confirms that the furniture sector is in a phase of transition and experimentation. Companies (and researchers) are making small prototypes: models that remain relatively close to the traditional and familiar business structures inspired by other sectors. However, the limited integration, especially of result-oriented PSSs, could be a symptom of the current limits in organizational capabilities, market acceptance, value chain coordination, and sector-specific knowledge. These topics need to be addressed in depth in the context of a sustainable and circular transition yet are beyond the scope of this research.

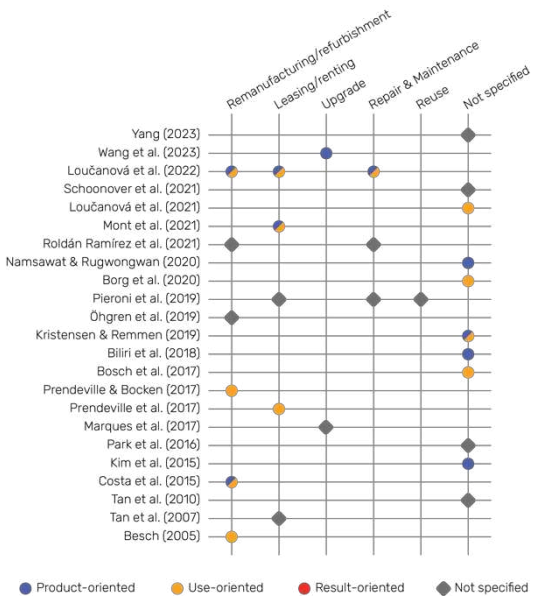

Figure 9: Sustainability Strategies Adopted Within the Three PSS Categories (Tukker 2004)

Discussion

The research results enable reflections on several levels. By critically examining the analytical characterization, it is possible to highlight how individual national landscapes are largely unexplored. For example, the Italian territory, although historically linked to design and the furniture industry, is devoid of contributions and experiments in the field of PSS. In fact, it was not possible to find articles reporting company initiatives nor studies highlighting user

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perceptions or barriers and enabling factors. Future research could cover these topics by initiating dialogues with companies and consumers and collaborating with academia, companies, and other organizations. The analytical characterization also shows that scientific journals historically linked to the field of design research are currently poor in articles on this topic, and future special issues on the subject are welcome. This absence may be due firstly to the disciplinary legacy of PSS and—as seen above—the closely related theme of sustainability and circularity, which is why different journals may have been prioritized. Secondly, the topic of PSS has very blurred boundaries. Indeed, it can be observed through the lens of sustainability, design, business models, or industrial systems. This multidisciplinarity generates a fragmentation of contributions that, depending on the focus, are published in one journal rather than another. Finally, practitioners still doubt the effectiveness of PSSs and servitization from an economical and practical implementation point of view. This is why design aspects may have been put aside and taken for granted.

From the thematic analysis of the contributions, the literature (and companies) have focused only on certain furniture products (such as office furniture and especially chairs) and have recently opened up to new types. As mentioned above, this focus may be due to both economic-productive reasons (such as the quantities required to make the efforts profitable), commercial reasons (the ease of creating contracts with specific types of organizations), and cultural reasons (resistance to change among both organizations and consumers). On the one hand, this narrow focus on specific furniture categories leads to a delay in the furniture industry's exploration of PSS solutions in B2B and (mainly) BC2 relationships, unlike in other sectors, such as in the case of transport with car-sharing or bike-sharing services. This focus can also be coupled with the most explored sustainability strategies—i.e., remanufacturing/refurbishment and leasing/renting, which are easily coupled and applicable to the B2B sector. On the other hand, the application of system innovations in the furniture sector presents specific challenges. These challenges are largely due to the characteristics of furniture products—with their physical, economic, and cultural characteristics. These factors make it more complex to implement circular and service-based models, requiring greater effort and commitment from companies. The furniture product and market peculiarities represent an opportunity for companies and researchers to explore how different product types can be linked to the concept of PSS, experimenting and exploring different strategies, new organizational structures, new business models, and collaboration between different stakeholders in the ecosystem.

As mentioned above, PSS are "new" models that, if properly designed, have the potential to drive system changes and, thus, sustainable transitions (Ceschin 2012). However, the current literature, at least in the furniture sector, is still to be investigated in depth. Indeed, most of the articles create connections and bridges between product and service aspects (also given the nature of PSS and the focus of the research) but do not explore with the necessary and detailed information how, for their implementation, companies have changed their

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organizational structures accordingly. Similarly, collaboration between different PSS stakeholders in the ecosystem is not explored in depth and only focuses on a single company or two within the value chain. Future research may explore how, through designing and developing PSSs, it is possible to create multi-level connections that lead to change at the organizational and product level yet are also able to change a sociotechnical system and influence the culture of an entire sector toward sustainability (Baldassarre et al. 2020).

Implications for Practitioners and Academics

The results obtained and previously discussed allow us to identify some implications and future directions for professionals such as designers, managers, researchers, and educators in design, sustainability, and PSS for the furniture sector. For design professionals and managers, the considerations drawn from the popularity of experiments in the contract furniture sector suggest that features such as a centralized supply, a strong supply chain, and a consolidated and contractualized relationship are fundamental for successful implementation of PSS. Understanding how to transfer these elements to consumers in the B2C sector without affecting the user experience and expectations could be successful, especially considering the integration of models such as renting/leasing or maintenance, repair, and upgrade. Furthermore, managers might need to involve other stakeholders (internal and external to the company) in developing new PSS offerings to increase awareness of individual ecosystems' characteristics and peculiarities and implement suitable strategies. Design-related tool, such as system maps, customer journeys, and service blueprints, can be applied in the project research and concept development phases precisely to identify latent needs and solve possible frictions in the experience. Moreover, the same tools can be used to examine the system required to implement the PSS, ensuring that all involved stakeholders have the opportunity to contribute smoothly and transparently. Companies can strategically emphasize product design strategies such as modularity, durability, and disassembly to support value-retention strategies such as remanufacturing and reuse.

For researchers and educators in the fields of design and PSS, the findings of this study highlight the need to strengthen multidisciplinary approaches—for instance, by connecting disciplines such as business management with design and by integrating well-established CE principles already developed in the literature, such as those by Tukker (2004), Bocken et al. (2016), Lüdeke-Freund et al. (2019), and Reike et al. (2018). Furthermore, from a field research perspective, it becomes important to understand how result-oriented PSS can be made more accessible and operational, particularly in emerging social contexts such as co-living, elderly care, and co-working—areas in which furniture can play a key role in enhancing and measuring performance. The research also points to the need to investigate geographical and cultural influences on adopting PSS, especially in Southern Europe and Asia. Finally, for researchers, there is an opportunity to develop long-term studies in collaboration with companies to assess

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the real-world effectiveness of PSS and evaluate how these models integrate within the systems and how all stakeholders change to support their implementation.

Conclusion

This research project mapped the academic state of the art on PSS designed to drive sustainable transitions in the furniture sector. Through a systematic literature review, twenty-three unique contributions were identified, analytically and thematically characterized. The analytical review revealed recent interest in this topic, particularly from Northern European academics. However, several data sources remain unexplored. For instance, the Italian context—encompassing users, businesses, culture, and the relationship with design—represents a valuable yet underutilized resource for facilitating PSS development. While most articles establish connections between products and services, the systemic changes (both organizational and external) required for their design and implementation are not adequately explored. Future research could investigate how PSS can create multi-level connections, prompting organizational and product-levels changes while influencing sociotechnical systems and cultural practices within sectors and communities to achieve sustainability goals.

The thematic analysis highlights how contributions fall into theoretical or practical categories, offering processes, methodologies, and tools for design; identifying barriers and enablers; capturing consumer perceptions; and reporting transition experiences. The office furniture sector dominates, with the most studied product- and use-oriented PSS. These are often paired with remanufacturing/refurbishment and leasing/renting strategies. From a practical perspective, the furniture sector remains heavily reliant on physical products, making natural resource consumption a critical issue. Adopting PSS could significantly reduce resource use, transforming products into revenue-generating assets that minimize environmental harm while meeting user needs. However, while extensively explored, product-, and use-oriented PSS still have higher environmental and resource costs compared to result-oriented PSS. Despite their challenges, lessons from other sectors could inform the design of result-oriented PSS for the furniture industry.

However, the analysis presented has limitations, first and foremost, the subjectivity in interpreting and categorizing the contributions. Although strategies have been adopted to limit the subjective element, such as the reading and interpretation by several authors, subjectivity is part of the design discipline and the interpretivist paradigm. By adopting a different paradigm or different methods of analysis (e.g., bibliometric mapping or quantitative content analysis), the selected contributions could be categorized differently. A second limitation might be the databases used to extract the contributions. By using different databases such as Google Scholar or Jstor, additional contributions could emerge that enrich or modify the results. Similarly, a further limit concerns the time frame, which was limited to contributions published up to mid-2024. Since PSS, sustainability, and CE are rapidly

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evolving topics, more recent publications may introduce further considerations. Likewise, the choice of specific keywords—while deliberately focused on capturing sector-specific knowledge about PSS—may have excluded useful contributions. Future research could update the review to include post-2024 publications and broaden the research strategy, testing new keyword combinations, and extracting data from different databases to get a more comprehensive picture of the furniture sector. Furthermore, future research could develop the results and considerations obtained through this research with additional data from the field, such as interviews with experts or PSS case studies, leading to a sector-specific conceptual model for furniture that links PSS models, sustainability/CE strategies, and industry characteristics.

In conclusion, PSS research is a holistic and fertile field. The relationship between sources and resources is continuous and inseparable, highlighting the necessary interplay between academia and practice. Academic resources can inform practical implementations, generating experiences and insights that enrich academic knowledge. To this end, the results of this study will be disseminated through reports to furniture companies to foster PSS implementation and facilitate sustainable transitions.

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The authors acknowledge the use of ChatGPT to integrate specific sources, partially define recommendations, and to improve the clarity and readability of the article. The prompts used include “This sentence is quite fragmented and hard to follow, can you help me to simplify it?”; “Can you help me find academic references regarding the first participatory design activities in Scandinavia, which began with a focus on office furniture?” or “Can you help me identify other implications of [...] for designers and design managers besides [....]?” The output from these prompts was used to facilitate reading, identify sources on a specific topic, and identify further implications. While the authors acknowledge the usage of AI, the authors maintain that they, Mattia Italia and Xue Pei, are the sole authors of this article and take full responsibility for the content therein, as outlined in COPE recommendations.

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Informed Consent

The authors declare that informed consent was not required as there were no human participants involved.

Conflict of Interest

The authors declare that there is no conflict of interest.

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Appendix

Table A1: Complete List of the Twenty-Three Selected and Analyzed Contributions and Their Keywords

No.	*Full Reference (APA Style)*	*Keywords of the Contribution*
01	Yang, D. (2023). Design for environmentally sustainable furniture systems—The knowledge and know-how of furniture life cycle design and furniture sustainable product-service system design. *The Design Journal, 26*(3), 503–514. https://doi.org/10.1080/14606925.2023.2178717.	Sustainable furniture system; design for sustainability; life cycle design; sustainable product-service system design
02	Wang, H., Mitake, Y., Tsutsui, Y., Alfarisi, S., & Shimomura, Y. (2023). A requirement analysis method for the design of the upgradable product-service system. *Procedia CIRP, 119*, 402–407. https://doi.org/10.1016/j.procir.2023.01.006.	Upgradable product-service system; customer requirement analysis; scenario building; upgrade strategy
03	Loučanová, E., Olšiaková, M., & Paluš, H. (2022). Consumers' perception of eco-services innovations related to furniture. *Drvna Industrija, 73*(4), 463–473. https://doi.org/10.5552/drvind.2022.2141.	Innovation; ecological innovation; services; consumers; furniture industry; Kano model
04	Schoonover, H. A., Mont, O., & Lehner, M. (2021). Exploring barriers to implementing product-service systems for home furnishings. *Journal of Cleaner Production, 295*, 126286. https://doi.org/10.1016/j.jclepro.2021.126286.	Product-service systems; leasing; furniture rental; home furnishings; business models
05	Loučanová, E., Olšiaková, M., Supin, M., & Drlickova, E. (2021). Ecological innovations in services—Servitization of furniture. In M. Jost (Ed.), *Woodema 2021: The response of the forest-based sector to changes in the global economy* (pp. 239–244). WoodEMA, i.a.—International Association for Economics and Management in Wood Processing and Furniture Manufacturing.	Innovation; ecological innovation; servitization; furniture; IKEA

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06	Mont, O., Lehner, M., & Schoonover, H. (2021). *Business models for sustainable consumption: Inspirational examples for the furniture and home textiles sectors* (Mistra Sustainable Consumption Report 1:9, p. 47). KTH Royal Institute of Technology.	—
07	Roldán Ramírez, B. M., Maya Duque, P. A., & Usuga Manco, O. C. (2021). Configuration of the product as a service strategy within the framework of Circular Economy, in the Furniture Sector of Higher Education Institutions. In *2nd South American International Conference on Industrial Engineering and Operations Management*, https://doi.org/10.46254/SA02.20210591.	Circular economy; product as a service; furniture; higher education institutions
08	Namsawat, O., & Rugwongwan, Y. (2020). An evaluation of corporate customer need with regard to the use of product service systems for the furniture business through environmental marketing. *Kasetsart Journal of Social Sciences, 41*(2), Article 2.	Corporate consumers; ecodesign; environmental marketing; furniture; product-service systems
09	Borg, D., Mont, O., & Schoonover, H. (2020). Consumer acceptance and value in use-oriented product-service systems: Lessons from Swedish consumer goods companies. *Sustainability, 12*(19), Article 19. https://doi.org/10.3390/su12198079.	Consumer acceptance; consumer perceived value; use-oriented product-service systems; consumer goods; Sweden
10	Pieroni, M. P. P., McAloone, T. C., & Pigosso, D. C. A. (2019). Configuring new business models for circular economy through product-service systems. *Sustainability, 11*(13), Article 13. https://doi.org/10.3390/su11133727.	Business model innovation; circular economy; product-service system; configuration; action research
11	Öhgren, M., Milios, L., Dalhammar, C., & Lindahl, M. (2019). Public procurement of reconditioned furniture and the potential transition to product service systems solutions. In G. Q. Huang, T. Qu, M. Thurer, S. Xu, & M. Khalgui (Eds.), *11th CIRP conference on industrial product-service systems* (Vol. 83, pp. 151–156). Elsevier. https://doi.org/10.1016/j.procir.2019.02.134.	Furniture reconditioning; public procurement; resource efficiency; circular economy; product-service systems
12	Kristensen, H. S., & Remmen, A. (2019). A framework for sustainable value propositions in product-service systems. *Journal of Cleaner Production, 223*, 25–35. https://doi.org/10.1016/j.jclepro.2019.03.074.	Product-service system; sustainable value proposition; value creation; circular economy; learning environment
13	Biliri, E., Lampathaki, F., Arvanitakis, A., Michalitsi-Psarrou, A., Martin, J., Gigante, F., Sales, V., & Nunez, M. J. (2018). Manufacturing intelligence in furniture product-service design. *Lecture Notes in Computer Science (Including Subseries Lecture*	Product-service; product-service system; manufacturing; sentiment analysis;

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	Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), 10697 LNCS, 246–250. https://doi.org/10.1007/978-3-319-73805-5_26.	knowledge management; workplace monitoring
14	Bosch, T., Verploegen, K., Grösser, S. N., & van Rhijn, G. (2017). Sustainable furniture that grows with end-users. In S. N. Grösser, A. Reyes-Lecuona, & G. Granholm (Eds.), *Dynamics of long-life assets: From technology adaptation to upgrading the business model* (pp. 303–326). Springer International Publishing. https://doi.org/10.1007/978-3-319-45438-2_16.	Circular economy; business models; life cycle analysis; system dynamics
15	Prendeville, S., & Bocken, N. (2017). Design for remanufacturing and circular business models. In M. Matsumoto, K. Masui, S. Fukushige, & S. Kondoh (Eds.), *Sustainability through innovation in product life cycle design* (pp. 269–283). Springer. https://doi.org/10.1007/978-981-10-0471-1_18.	Design for remanufacturing; ecodesign; eco-innovation; sustainable business models; circular economy
16	Prendeville, S., O'Connor, F., Bocken, N. M. P., & Bakker, C. (2017). Uncovering ecodesign dilemmas: A path to business model innovation. *Journal of Cleaner Production, 143*, 1327–1339. https://doi.org/10.1016/j.jclepro.2016.11.095.	Ecodesign; ecodesign maturity; dilemmas; trade-offs; business model innovation
17	Marques, M., Poler, R., Agostinho, C., & Jardim-Goncalves, R. (2017). An architecture to support the development of reconfigurable and updatable product-service systems in furniture sector. *2017 International Conference on Engineering, Technology and Innovation (ICE/ITMC)*, 1076–1081. https://doi.org/10.1109/ICE.2017.8280001.	Product-service systems; reconfigurability and updatability; industrial mass-customization
18	Park, Y., Kim, M., & Yoon, J. (2016). Generating new product-service system concepts using general needs and business system evolution patterns: A furniture PSS case. *Industrial Engineering and Management Systems, 15*(2), 181–195. https://doi.org/10.7232/iems.2016.15.2.181.	Product-service system; concept generation; general needs; system evolution patterns; furniture
19	Kim, Y. S., Lee, J., Lee, H., & Hong, Y. S. (2015). Product-service business concept design: Real-world case of a small furniture manufacturing firm. In X. Boucher & D. Brissaud (Eds.), *7th industrial product-service systems conference—IPSS, industry transformation for sustainability and business* (Vol. 30, pp. 257–262). Elsevier Science Bv. https://doi.org/10.1016/j.procir.2015.03.103.	Product-service business concept; servitization; product-service systems design; business model
20	Costa, F., Prendeville, S., Beverley, K., Teso, G., & Brooker, C. (2015). Sustainable product-service systems for an office furniture manufacturer: How insights from a pilot study can inform PSS design. *Procedia CIRP, 30*, 66–71. https://doi.org/10.1016/j.procir.2015.02.109.	Sustainable PSS; service design; LCA; pilot study; ECO-WISE

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21	Tan, A. R., Matzen, D., McAloone, T. C., & Evans, S. (2010). Strategies for designing and developing services for manufacturing firms. *CIRP Journal of Manufacturing Science and Technology*, *3*(2), 90–97. https://doi.org/10.1016/j.cirpj.2010.01.001.	Product/service-systems; design for X; product life cycle design; service design
22	Tan, A. R., McAloone, T. C., & Gall, C. (2007). Product/service-system development—An explorative case study in a manufacturing company. *DS 42: Proceedings of the 16th international conference on engineering design (ICED 2007)*, Paris, France.	Product/service-systems; innovation strategy; sustainability; workspace; value networks
23	Besch, K. (2005). Product-service systems for office furniture: Barriers and opportunities on the European market. *Journal of Cleaner Production*, *13*(10), 1083–1094. https://doi.org/10.1016/j.jclepro.2004.12.003.	Product-service systems; office furniture; product lifetime extension; remanufacturing; PSS scenario

Table A2: Duration and Partnership of Articles That Made a Scientific Contribution Through Action Research.

Contribution	*Project Duration*	*Partner Company*
Bosch et al. (2017)	2013–2016 (UIW Project)	Gispen
Prendeville and Bocken (2017)	2015–2017 (European Remanufacturing Network Project)	Not specified
Prendeville et al. (2017)	Not specified	Orangebox
Costa et al. (2015)	2014–2016 (Office Chair for Life Project—Innovate UK Funded[2])	Orangebox
Tan et al. (2007)	Not specified	Steelcase

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[2] Innovate UK Projects: https://www.ukri.org/publications/innovate-uk-funded-projects-since-2004/.

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